

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 July 2026
Name	Mallampati Bharadwaj
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

Component	Description	Technology
User Interface	Service Catalog interface where users submit network resource requests by selecting device_type, connection_type, address, and user_name.	ServiceNow Service Portal, Service Catalog, HTML, CSS, JavaScript
Application Logic-1	Validates catalog inputs and triggers the automation workflow after request submission.	ServiceNow Flow Designer, Catalog Client Scripts, UI Policies
Application Logic-2	Handles approval workflow, decision logic, and conditional branching (Approved/Rejected).	Flow Designer, Approval Engine, Business Rules
Application Logic-3	Inserts approved request details into the custom backend table and sends notification emails.	Flow Designer Actions, Script Includes, Notification Engine
Database	Stores all approved network resource request records.	ServiceNow Custom Table (u_network_database)
Cloud Database	Cloud-hosted enterprise database managed by the ServiceNow platform.	ServiceNow Cloud Database (Now Platform)
File Storage	Stores attachments such as network diagrams or supporting documents submitted with requests (if enabled).	ServiceNow Attachment Management
External API-1	Optional integration with enterprise systems (Asset Management, CMDB, or ITSM tools).	ServiceNow REST API
External API-2	Optional integration for email notifications or third-party network management systems.	SMTP Email Service, REST/SOAP APIs
Machine Learning Model	Intelligent request classification, approval prediction, or anomaly detection (future enhancement).	ServiceNow Predictive Intelligence (Optional)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	ServiceNow Now Platform (ITSM Framework), Glide Framework, AngularJS (Service Portal), Bootstrap, jQuery
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	SHA-256 Encryption, HTTPS/TLS, Role-Based Access Control (RBAC), ServiceNow ACLs, Multi-Factor Authentication (MFA), Audit Logging, OWASP Top 10 Compliance
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	ServiceNow cloud-native multi-tenant architecture with horizontal scalability, Flow Designer for distributed workflow execution, micro-services based platform
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	99.9% platform uptime (ServiceNow SLA), Load Balancing, High Availability Clusters, Automated Failover, Redundant Infrastructure
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Flow Designer optimized actions, GlideRecord query optimization, Business Rule best practices, Caching (GlideCache), CDN for static content, Asynchronous processing for high-volume requests